

Rohit Suryaa Saravanan

Software Developer and Data Scientist

Fulda, Hessen, Germany · +49 160 96597803 · Rohitsuryaas21@gmail.com
linkedin.com/in/rohitsuryaa · rohitsuryaa.com · github.com/srohitsuryaa21



SUMMARY

Software developer and data scientist, currently doing an MSc in Data Science at Hochschule Fulda. One year of professional experience building enterprise software for a US cybersecurity startup, two IEEE papers in applied machine learning, and practical work on AI agents built with LangGraph and RAG. I care about the model, and just as much about the API around it, the pipeline that ships it, and whether the result actually helps.

EXPERIENCE

Associate Software Developer · EigenSecure, remote (USA) Mar 2024 to Apr 2025

- Turned business and security requirements into backend features, REST APIs and proof of concept components in an Agile Scrum team.
- Built features around access control, evidence workflows and compliance data, working across MySQL, PostgreSQL and MongoDB.
- Improved SQL queries, data workflows and CI/CD pipelines for reliable deployments and traceable audit trails.
- Kept engineering and product stakeholders informed about progress and the choices involved, balancing security needs with delivery speed.

SELECTED PROJECTS

Windmill Pitch Prediction · IEEE ICEMCE 2023 Feb to Jun 2024

- Built 26 time based features from six months of turbine data, trained an XGBoost model and added a second layer that corrects its remaining errors.
- Reached 96% prediction accuracy, cut error from 8.6% to 2.3% and raised modelled power output by 21%.

Stack: Python, XGBoost, time series, simulation, statistical evaluation · DOI 10.1109/ICEMCE57940.2023.10433983

Supply Chain Intelligence · Forecasting and analytics pipeline 2025

- Built an analytics pipeline over 110,123 online orders: SARIMA and XGBoost forecasting with error under 20% (MAPE 18.5%), EOQ reorder quantities, delivery analysis at 92.1% on time and risk scores for 881 sellers.
- Brought the results together in one interactive Plotly Dash dashboard for operational decisions.

Stack: Python, Pandas, NumPy, SARIMA, XGBoost, SQL, Plotly Dash · github.com/srohitsuryaa21/supply-chain-forecasting

Luxury Concierge Agent · AI agent for vehicle configuration 2026

- Built a five step LangGraph agent over a SQLite catalogue that turns usage, region, cabin style, budget and timeline into a priced specification ready for sales.
- Every answer is checked against the catalogue with Pydantic, budgets are enforced in Python so the model can never set a price, conversations are saved, and FastAPI serves a hand written front end. Twenty offline test scenarios run without a single API call.

Stack: Python, LangGraph, RAG, FastAPI, Pydantic, SQLite, pytest · github.com/srohitsuryaa21/luxury-concierge-agent

FinGuard PayOps · Payment operations platform

- Built a SEPA style payment lifecycle with rule based fraud scoring, compliance alerts, an approve or reject flow for analysts, audit logging and a React dashboard, designed with DORA style resilience in mind.

Stack: Java, Spring Boot, PostgreSQL, React · github.com/srohitsuryaa21/FinGuard-PayOps

DeFi Farmer Platform · IEEE ICSCSS 2024 2024

- Wrote Ethereum smart contracts for loans, payments, insurance and supply chain tracking for small farmers, with a Web3.py backend tested on a local Ganache network. Modelled adoption of digital records rose from under 50% to about 80%.

Stack: Solidity, Ethereum, Web3.py, Ganache · DOI 10.1109/ICSCSS60660.2024.10625462

Aviation Delay Ontology · Data pipeline and Power BI

- Modelled 63,000 US flights as connected flights and airports in the style of Palantir Foundry: cleaning, delay features, stable flight IDs and an airport table in pandas and PySpark, reported in Power BI (Chicago O'Hare averaged 21.1 minutes of delay against 13.6 at Atlanta).

Stack: Python, pandas, PySpark, Power BI · github.com/srohitsuryaa21/aviation-delay-ontology

- Estimated battery deployment from speed and throttle in public FastF1 data for the 2026 British Grand Prix and measured tyre wear per stint; the two turned out to be unrelated ($r = -0.04$).

Stack: Python, FastF1, statistics · github.com/srohitsuryaa21/SOC-Silverstone-2026

Employee Attrition Prediction · Explainable machine learning

Dec 2022 to Feb 2023

- Classification pipeline on a public HR dataset with exploratory analysis, feature engineering, model comparison and driver analysis that explains which factors matter most.

Stack: Python, Scikit-learn, Pandas · github.com/srohitsuryaa21/employee-attrition-prediction

Personal portfolio · rohitsuryaa.com

2025 to now

- Bilingual website in English and German with scroll driven storytelling, a canvas data visualisation and full fallbacks for reduced motion and screen readers.

Stack: Astro, TypeScript, GSAP, CSS · rohitsuryaa.com

EDUCATION

MSc Data Science · Hochschule Fulda, Germany

Apr 2025 to now

Machine learning, big data and distributed systems, data mining and predictive analytics, process mining, visualisation and BI, research methods, data ethics and GDPR.

BTech Big Data Computer Science · SRM University, Chennai, India

Aug 2020 to Nov 2024

Big data systems, distributed systems, data structures and algorithms, databases, software engineering.

TECHNICAL SKILLS

Data and ML	Python, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, TensorFlow, SARIMA, time series, classification, explainability, statistics
AI agents	LangGraph, RAG, FastAPI, Pydantic, ChromaDB, Ollama, pytest, offline evaluation
Software	C#, .NET, ASP.NET Core, Blazor, Java, Spring Boot, REST APIs, JavaScript, TypeScript, React
Data and BI	SQL, PostgreSQL, MySQL, MongoDB, SQLite, PySpark, ETL, Power BI, Plotly Dash, Tableau
Cloud and delivery	Azure, Docker, GitHub Actions, Jenkins, CI/CD, Git, Linux

PUBLICATIONS

- Windmill pitch prediction with XGBoost and residual correction. IEEE ICEMCE, 2023. DOI 10.1109/ICEMCE57940.2023.10433983
- A DeFi platform for transparent agricultural finance. IEEE ICSCSS, 2024. DOI 10.1109/ICSCSS60660.2024.10625462

CERTIFICATIONS

LANGUAGES

IBM Data Science Professional (Coursera)

English: fluent (C1+) · German: A2, improving · Tamil: native